Nova Information Management School
BSc in Data Science
**Text Mining 2024-2025**

## Group Project: *"Solving the Hyderabadi Word Soup"*

# Project Report


**Group 11**

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## 1. Introduction

Who has never regretted going to a restaurant? Whether it's the food, the ambiance, or the service, our dining experiences are influenced by a variety of factors that often end up in the form of online reviews.

This report dives into a dataset of restaurant reviews and metadata to uncover patterns, predict restaurant ratings, and analyze sentiments behind customer feedback, offering structured approaches to solve complex problems. Our analysis is focused on three tasks: predicting restaurant cuisines through multilabel classification (IR 3311); analyzing sentiment to predict restaurant ratings (IR 3312); and clustering dishes based on co-occurrence patterns in the reviews (IR 3321). The final requirement involving topic modelling (IR 3322) was not explored.

The datasets that were given to us were explored, the defined cleaning pipeline applied given the needs of each task, before implementing and evaluating models, each tailored to address specific research objectives. The results provide valuable insights into the datasets, revealing patterns and successfully inferring about restaurant ratings and cuisines.

## 2. Literature Review

The analysis of restaurant reviews has become a popular domain of study due to the growing influence of online feedback in shaping consumer decisions. Previous research has shown that restaurant ratings and reviews often serve as a reliable quality indicator, with aspects like food, service, pricing, and ambiance playing significant roles in customer satisfaction (Gan et al., 2015). [1]

Moreover, multilabel classification techniques of reviews allow for more nuanced predictions compared to traditional single-label classification, capturing the diversity of object of study offerings (restaurants cuisines, in our case). These approaches have already been tested and their specificities addressed (Singh et al., 2022). [2]

Co-occurrence analysis and clustering have also been used to identify patterns in consumer preferences. Techniques like clustering and named-entity recognition (NER) have been employed to deal and distinguish between foods, ingredients, and dishes, grouping them; and providing insights into the most popular or emerging food trends (Diwan et al., 2020). [3]

This report builds upon these methodologies, extending them to our specific tasks, that were mainly performed based on class materials.

## 3. Data Understanding

Main insights:

- Reviews' and restaurants' ratings follow very different distributions, with a left-skewed and a close-to-normal distribution, respectively. Nonetheless, the average rating of both reviews and restaurants is 3.6 stars;

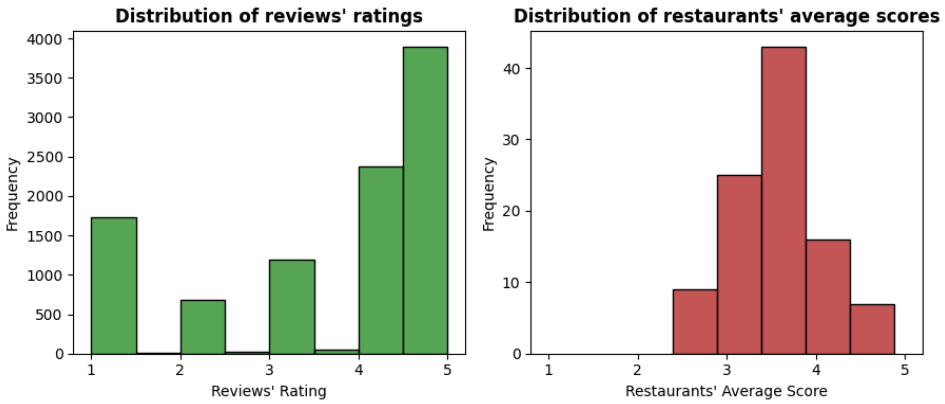


Fig. 1 - Distribution of reviews' and restaurants' ratings

- Although the reviews date from 2016, the majority come after the second half of 2018;

- The normal price of a meal is around 700 rupees;

- Restaurants' meal costs and average rating are moderately correlated;

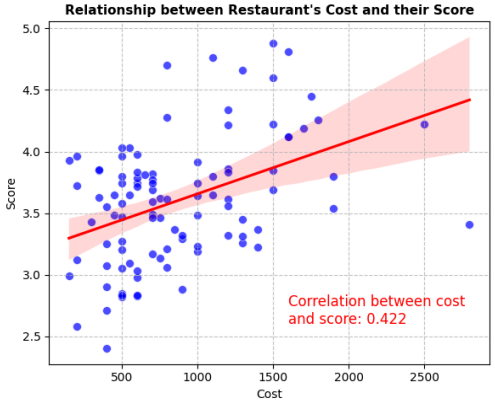


Fig. 2 - Scatterplot of the relationship between restaurants' cost and scores

- The most frequent cuisines are North Indian and Chinese, which have an average rating;

- All the most expensive cuisines tend to have close to average ratings, except Mediterranean, which is the best-rated cuisine by far;

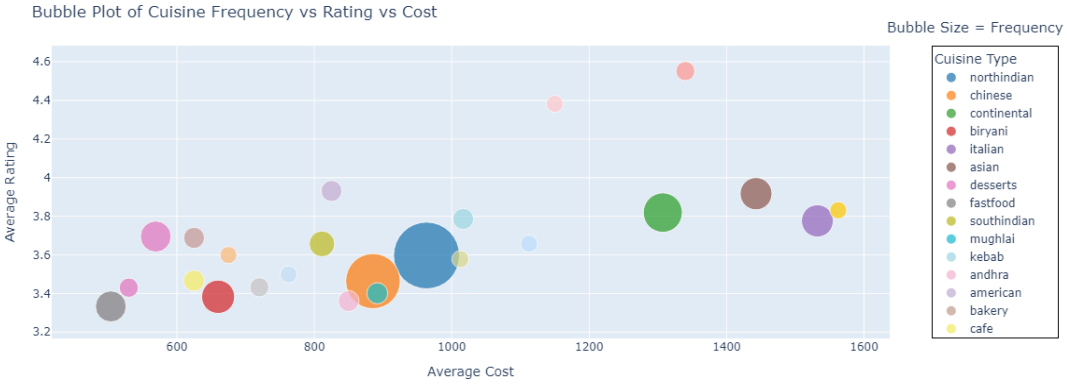


Fig. 3 - Bubble plot of cuisine frequency, rating, and cost

- The reviews' word cloud tells us that there is an overall positive sentiment in the reviews, and the food itself is not the only decisive factor in reviews, with words like "ambience", "service", and "staff" being very frequent.

Fig. 4 - Word cloud of the entire dataset

- Two very similar cuisines - North and South Indian, have similar vocabulary, as expected. Nonetheless, biryani and paneer dishes seem to be more common in the north, whereas vegetarian dishes are more regular in the south.

Fig. 5 - Word clouds of the reviews of North Indian (left) and South Indian (right) cuisine restaurants

# 4. Data Preparation

## 41. General Preprocessing

This stage is crucial to ensure the dataset is reliable and ready for analysis, by cleaning the data and correcting inconsistencies. These operations were done before data understanding so that the datasets are workable and processable.
Main operations:

- Cleaned one record on *reviews* that had a 'Rating' of 'Like', instead of a number 0-5;
- Updated the 'Time' variable data type to datetime and 'Rating' to numerical on the *reviews* dataset;
- Excluded every record with missing values on *reviews* (only 0.5% of them), as any of them had the review text, a vital part for every information requirement;
- Cleaned the 'Cost' variable of *restaurants* and set its datatype to numerical;
- Created a 'Score' variable on *restaurants* to store the average rating of each restaurant;
- Transformed the 'Cuisines' variable of *restaurants*, with each restaurant having a list of their cuisines, in a more processable way.

## 42. Additional Data Cleaning & Transformations

In this part, there are the operations that were not crucial for performing the data understanding part, but still very much needed to clean and facilitate our data.

Main operations:

- Updated the column data types on *reviews;*
- Converted the 'Metadata' variable of *reviews* into two different features containing the reviews and followers;
- Dropped the 'Link', ‘Collections’, and ‘Timings’ of *restaurants*;

- In the end, we integrated the reviews dataset on the restaurants' as most of the information requirements are restaurant-based.

### 43. IR 3311 Data Preparation

Main steps:

- Cleaned and prepared the raw review's text, by applying the default cleaning pipeline;
- Resolved the issue of having very rare cuisines that previously prevented the use of cross validation. That was done by assigning cuisines to the ones that they had co-occurred the most, and in case of ties, a GloVe pre-trained model [4] was leveraged to give us the best option of assignment;
- Binarized our target variable, the cuisine type;
- Split the data into a train and test set: stratified k-fold cross validation would be applied in the training set, to robustly train the models; and an isolated test set was also crucial to avoid data leakage, on which models will be evaluated.

### 44. IR 3312 Data Preparation

Main steps:

- Cleaned and prepared the raw review's text, by applying the cleaning pipeline specifically adapted to a sentiment analysis task (keeping stopwords, uppercase, and punctuation);
- It was applied with three different approaches: to the entire reviews of a restaurant; to every sentence on a review; and to every paragraph on a review. These will be tested and compared against each other, to choose the best approach.

### 45. IR 3321 Data Preparation

Main steps:

- Cleaned and prepared the raw review's text, by applying the cleaning pipeline specifically adapted to a named-entity recognition task (keeping stopwords, punctuation), but opted to lowercase the entire reviews, as our goal is to extract only dishes. If the goal was to extract, for instance, countries or companies, it would not be a good practice in a NER task;
- Tokenized and extracted POS with the spacy's *en_core_web_sm* model;
- Integrated an ingredient dataset from Kaggle, to later efficiently distinguish foods/ingredients from real dishes.

## 5. Modelling

### 51. IR 3311 Model Implementation

Main steps:

- Defined a customizable classifier, that allows using and testing of distinct preprocessing, vectorizing, and classifying techniques. This is essential to avoid data leakage, processing each set independently;
- Performed a small test on the classifier with K-fold cross validation;
- Performed a Grid Search to find the combination of parameters that yields the best performance;
- Found the best model, which was the one with the fastest fitting and scoring time, among the 16 top models on the performance metrics with a tie.

## 52. IR 3312 Model Implementation

Main steps:

- Generated polarity scores of all approaches used using both Vader and TextBlob. The different approaches with Vader yielded distinct distributions, while TextBlob gave similar distributions independently of the approach.

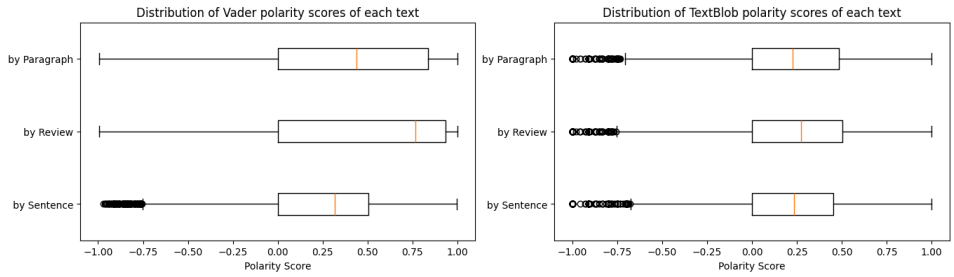


Fig. 6 - Distributions of polarity of the different approaches tested

- Compared the normalized scores to the target variable. No approach had a very close distribution to the normalized restaurant score (blue), but everyone (except Vader by Review) had similar shapes but with a higher kurtosis.

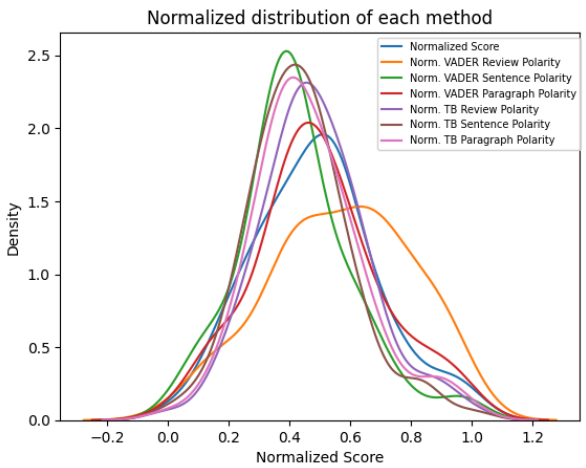


Fig. 7 - Comparison of normalized distributions of every approach to restaurants' scores

- The chosen approach was TextBlob by Review, based on visual inspection and an undisputedly best performance on selected metrics (RMSE and MAPE).

- Finally, we developed a linear regression model, to make the bridge between the polarity extracted and the restaurants' scores, in order to use the first to predict the latter.

### 53. IR 3321 Model Implementation

Main steps:

- Initially, a pre-trained BERT model [5] tailored for food extraction was tested, but it did not lead to good results;
- Then, a rule-based approach was applied, based on n-grams of tokens and POS, to extract dish names, which led to a better outcome;
- Specificities like misspellings, ingredients in the dish list, singular/plural redundancies, and x-word dishes were dealt with;
- With a final dish list, a co-occurrence matrix was created, and later transformed into a distance matrix, that served as input to the OPTICS clustering algorithm, which outperformed HDBSCAN in our tests;
- Finally, the most frequent cuisine types from each cluster were computed and extracted.

## 6. Evaluation

### 61. IR 3311 Model Evaluation

The final model outperformed a dummy classifier in every global metric, and also in almost every cuisine type, after measuring their performance in the isolated test set. It scored over 0.5 in precision, recall and f1 weighted metrics. As expected, the cuisine types with more samples are better rated.

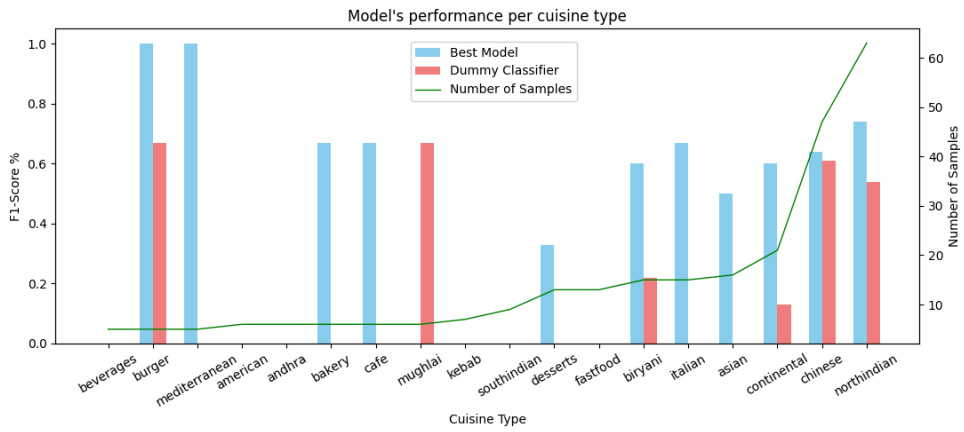


Fig. 8 - Comparison of F1-scores of the best model, and a dummy classifier, per cuisine type

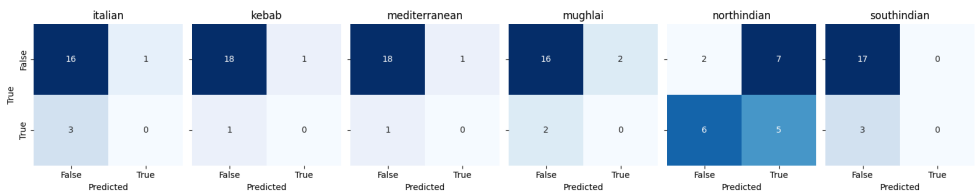

Fig. 9 - Confusion matrices of the final model's performance on the test set

Answer to the IR: The multilabel classification model performed well for majority cuisine types such as North Indian and Chinese, achieving satisfactory results due to their frequency in the dataset. However, for minority cuisine types, the model's performance was limited, indicating bias towards the most common cuisines. Despite this limitation, the model outperformed a dummy classifier across all global metrics and demonstrated its robustness with stratified k-fold cross-validation..

## 62. IR 3312 Model Evaluation

The final model has an excellent fit over the points, as expected since the two variables are almost perfectly correlated (0.936 Pearson's correlation). There are no positive/negative outliers in one variable, that also are outliers of the opposite sign in the other variable, another sign of excellent model performance. It also has very good results in the selected performance metrics. For instance, it has a MAPE of 0.038 and 0.046 in the training and test set, respectively. These scores show us that the average difference between predicted and real values is only around 4%.

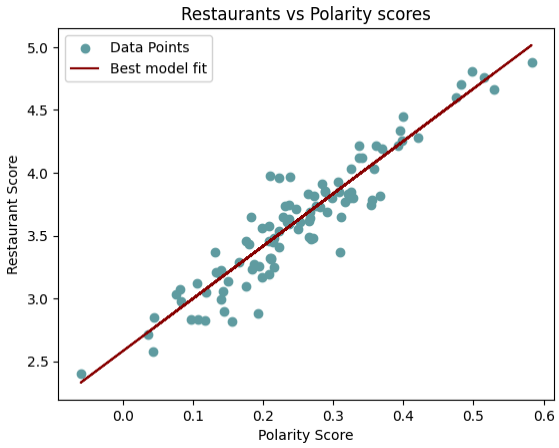


Fig.10 - Best fit of the model on the review's text

Answer to the IR: The polarity scores correlated closely with the restaurant's scores, indicating that review sentiment is a strong predictor of restaurant ratings. A simple regression model was built to quantify this, and its predictions have a mean error of only about 4%, indicating high precision in predicting restaurants' scores. Given this, we can confidently say that we are able to predict a restaurant's Zomato score using the polarity of their reviews as input very well.

## 63. IR 3321 Model Evaluation

The majority of the most frequent dishes are composed of chicken, with '*chicken wings*', '*chicken tikka*', and *'chicken biryani'* highlighted.

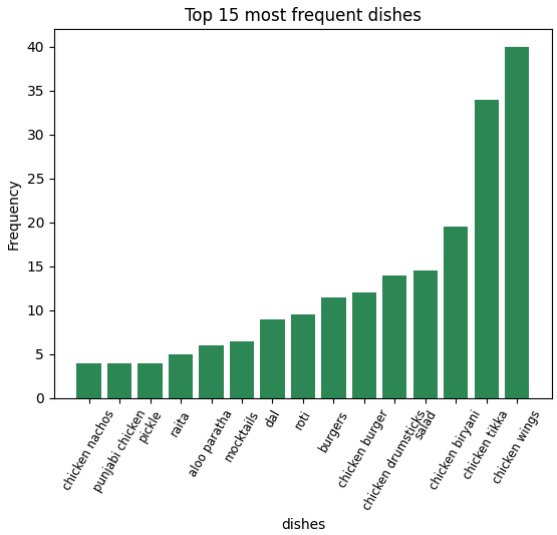


Fig.11 - Bar plot of the most frequent extracted dishes

As the majority of these restaurants are of north indian cuisine, most of the dishes are of the same origin. Given this, there is one big (northindian_chinese_biryani) cluster containing most of dishes, such as *paneer tikka, chicken curry*, and some less related like *chicken wings*, revealing not efficient clustering; Nonetheless, dishes from bakery (yellow) and dessert (purple) cuisines are well separated from the rest, with dishes as *cakes, waffles,* and *pancakes* as evidence; chinese (blue) and north indian (green) dishes, like *noodles*, or *roti* are also separated from the rest.

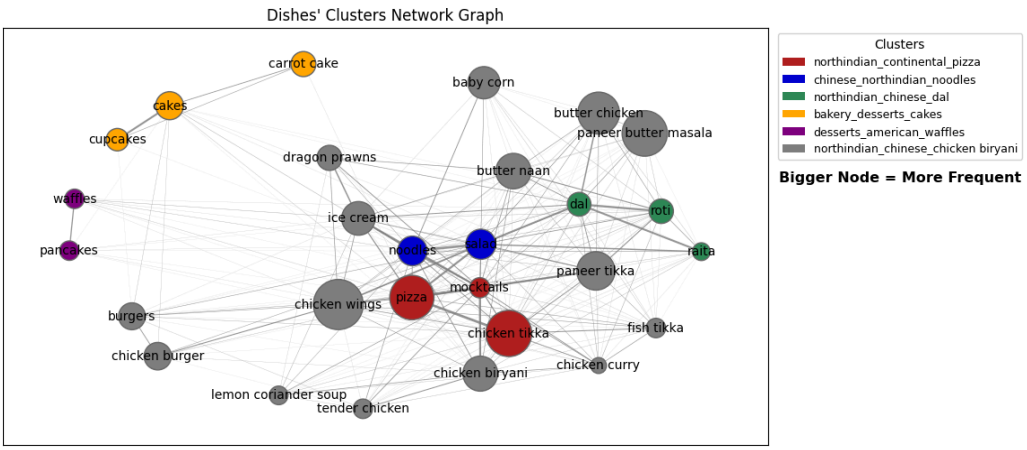


Fig.12 - Network graph of the final dish clustering solution, with cuisine types

<u>Answer to the IR:</u> Dishes from similar cuisine types tend to be mentioned together in the reviews. With North Indian being the majority cuisine in the dataset, most dishes are from there, which happens to bias the model to some degree. However, clusters with Bakery, Desserts, Chinese, and Continental cuisines were still able to be composed, all with specific dishes of that same cuisine.

## 7. Conclusions

Overall, this report addressed the explored information requirements in a satisfactory way. The methods used provided valuable insights into the factors influencing restaurant reviews: the multilabel classification model effectively predicted major cuisine types like North Indian and Chinese, while sentiment analysis demonstrated a strong correlation between review sentiment and restaurant ratings. Co-occurrence clustering revealed patterns in dish mentions, particularly among specific cuisines like Bakery and Chinese.

However, there are limitations in the chosen methods and data. The multilabel classification model showed satisfactory results for majority cuisine types but struggled with less frequent ones, indicating a bias toward popular cuisines; and the co-occurrence clustering was influenced by the dominance of North Indian cuisine in the dataset, which skewed results for other cuisines.

Overall, the report provides a robust analysis of the dataset and meets the core objectives, though future improvements could include balancing the dataset to reduce bias (or not, since the dataset is inherently from an Indian city), optimizing the multilabel classifier and the clustering methods to better account for rare and diverse cuisines.

## 8. References


Carina Albuquerque, Artur Varanda - Text Mining course materials

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[3]: N. Diwan, D. Batra, and G. Bagler, "A Named Entity Based Approach to Model Recipes," in *Proc. IEEE 36th Int. Conf. Data Eng. Workshops (ICDEW)*, 2020. Available: https://ieeexplore.ieee.org/document/9094103

[4] GloVe pre-trained model https://huggingface.co/fse/glove-wiki-gigaword-50

[5] BERT pre-trained NER model https://huggingface.co/Dizex/FoodBaseBERT-NER